

Lesson 22 Inverse Functions and Function Composition

In Lesson 21, we showed visually that the inverse relation f^{-1} is a function if and only if f is a bijection. We now give a proof and a definition.

Theorem Let $f: A \rightarrow B$. The inverse relation f^{-1} is a function from B to A if and only if f is a bijection.

Proof

($\Leftarrow$) Suppose f is a bijection, we will show that f^{-1} is a function from B to A . Let $y \in B$. Since f is onto, $(x, y) \in f$ for some $x \in A$, so $(y, x) \in f^{-1}$ for some $x \in A$. Suppose $(x, y) \in f^{-1}$ and $(x, z) \in f^{-1}$, then $(y, x) \in f$ and $(z, x) \in f$. Since f is one-to-one, $y = z$. So, every element in B is mapped to exactly an element in A under f^{-1} , and f^{-1} is a function from B to A .

($\Rightarrow$) Suppose f^{-1} is a function from B to A , we will show that f is one-to-one and onto. Suppose $(x, z) \in f$ and $(y, z) \in f$, then $(z, x) \in f^{-1}$ and $(z, y) \in f^{-1}$. Since f^{-1} is a function, then $x = y$. So f is one-to-one. Let $y \in B$. Since f^{-1} is a function from B to A , $(y, x) \in f^{-1}$ for some $x \in A$. So, $(x, y) \in f$ for some $x \in A$, and f is onto.

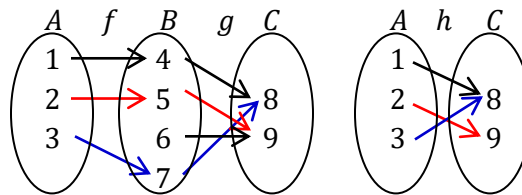
Proposition 1 If a function f is invertible, then f^{-1} is invertible and $(f^{-1})^{-1} = f$.

Proof If f is invertible, then f^{-1} is a function by definition. Since a function is a relation, we have $(f^{-1})^{-1} = f$ (as relations). This means that the inverse relation of the function f^{-1} is a function, namely f . So f^{-1} is invertible, and the inverse function of f^{-1} is f .

Next, we introduce a way of combining two functions to make a new function.

Definition Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions, then the **composition** of f and g is the function $g \circ f: A \rightarrow C$, where $(g \circ f)(x) = g(f(x))$.

Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ are defined as follows, then the composition of f and g is a function $h: A \rightarrow C$, which can be found as follows:



The new function h maps an element from A to an element in C , as follows:

$$\begin{aligned} h(1) &= g(f(1)) = g(4) = 8, & h(2) &= g(f(2)) = g(5) = 9, \\ h(3) &= g(f(3)) = g(7) = 8 \end{aligned}$$

Consider the real functions $f(x) = x^2 + 1$, and $g(x) = 2x - 3$. Then

$$(f \circ g)(x) = f(g(x)) = f(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 10$$

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = 2(x^2 + 1) - 3 = 2x^2 - 1$$

In general function composition is *not* commutative: $f \circ g \neq g \circ f$. On the other hand, composition is associative: $h \circ (g \circ f) = (h \circ g) \circ f$.

Theorem 2 Let $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ be functions, then

$$h \circ (g \circ f) = (h \circ g) \circ f$$

Proof Note that $\text{dom}(g \circ f) = \text{dom}(f)$ and the codomain of $g \circ f$ is the codomain of g , so

$$\text{dom}(h \circ (g \circ f)) = \text{dom}(g \circ f) = \text{dom}(f) = \text{dom}((h \circ g) \circ f)$$

$$\text{codomain}(h \circ (g \circ f)) = \text{codomain}((h \circ g) \circ f) = \text{codomain}(h)$$

Finally, let x be an arbitrary element of A , then

$$h \circ (g \circ f)(x) = h((g \circ f)(x)) = h(g(f(x)))$$

$$(h \circ g) \circ f(x) = ((h \circ g)(f(x))) = h(g(f(x)))$$

Proposition 3 Let A, B, C be sets, and suppose $f: A \rightarrow B$ and $g: B \rightarrow C$.

a) If f and g are one-to-one, then $g \circ f$ is one-to-one.

Proof Suppose $(g \circ f)(x) = (g \circ f)(y)$. Then $g(f(x)) = g(f(y))$.

Since g is one-to-one, we have $f(x) = f(y)$.

Since f is one-to-one, we have $x = y$.

b) If f and g are onto, then $g \circ f$ is onto.

Proof Let $z \in C$. Since g is onto, there exists $y \in B$ such that $g(y) = z$.

Since f is onto, there exists $x \in A$ such that $f(x) = y$.

So, there exists $x \in A$ such that $(g \circ f)(x) = g(f(x)) = g(y) = z$.

Proposition 3 says that the composition of two bijections f and g is a bijection, and in an (optional) exercise, we will prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.